The relationship between sleep and physical activity across the lifespan

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Abstract

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline.

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Introduction

Sleep is an important determinant of population health [1]. The consequences of poor sleep are profound and include increased all-cause mortality [2], cardiovascular disease [3], cancers [4], and poor mental health [5]. Poor sleep is common: over 50% of children and adolescents [6] and 35% of adults [7] report lower sleep duration than recommended. Improving sleep should be a priority for public health policy and more safe and accessible strategies that promote sleep are needed [8]. In adults, being physically active appears to improve both the quantity and quality of sleep [9] and the World Health Organisation recommends physical activity as a method to improve sleep in adults [10]. In children, adolescents, and older adults associations are unclear [8,11,12] and guidelines do not currently support a role for physical activity to improve sleep.

Promoting physical activity may be an effective method to improve sleep across the lifespan but the published literature to date has at least two limitations that prevent firm conclusions to support health guidelines. First, most studies have relied on self-report measures of sleep and physical activity [8,11,12], which could confound associations and limit the precision of analyses designed to identify a dose-response relationship between physical activity and sleep. Second, no study has collected data from a large sample that included participants across the lifespan. Umbrella reviews [8] and meta-analyses [3,4,9,11–28] have provided some suggestive evidence about how physical activity as a means of improving sleep may vary across the lifespan. However, their reliance on aggregated study-level effects limits the quality and scope of data that can be used to analyse the influence of age [29]. Individual participant data (IPD) analysis can overcome these limitations [30].

In this study we harmonised individual participant data (IPD) from activity monitor observations obtained from studies around the world. The objective of this pooled IPD study was to investigate the relationship between device-measured physical activity and sleep across the lifespan. We examined whether individuals who are more physically active have better sleep, whether age, sex, and socio-cultural factors moderate the association between physical activity and sleep, and whether there is a bidirectional association between physical activity and sleep.

Methods

We pre-registered all methods and analyses on the Open Science Framework (OSF) before data acquisition began (https://osf.io/ahd96).

Data sources and data acquisition

We included data from studies, with any design, that used activity monitors and a 24-hour wear time protocol over a minimum of 4 days, including 1 weekend day. Studies needed to have used research-grade accelerometers from which raw accelerations (m/s^2 or mg) could be obtained. Studies could use any wear location for the accelerometer (hip, wrist, or thigh).

We identified potentially relevant studies by examining the reference lists of recent systematic reviews that explored the relationship between sleep and physical activity in children, adolescents, and adults [9,11,12]. We then used forward searching on potentially eligible studies [31]. We additionally performed a literature search to identify accelerometry consortiums and individual studies, searched trial registries (ANZCTR, ICTRP, ISRCTN) and the Open Science Framework for unpublished results, and we asked collaborators to help identify any additional studies.

We contacted chief investigators or consortium leads by email to request access to deidentified individual data. To make the data acquisition manageable, we only invited studies with sample sizes larger than 400 participants. If required by the contributing research group, the contributing investigators or their university signed a data transfer agreement. All studies included in the pooled data meta-analysis had to have been approved by their local institutional ethical committees, and all participants had to have provided written informed consent. The lead university's Research Ethics Committee granted permission for each study's data to be obtained, and investigators transferred their deidentified data via a secure file-sharing system.

Data harmonisation and outcomes

All activity monitor data were harmonised by reprocessing raw accelerometer files using an open-source GGIR package in R to produce physical activity and sleep outcomes [32,33]. By reprocessing all raw accelerometer data, we were able to generate outcomes that could be compared across studies, devices wear location, and populations [34]. Data were reprocessed by the core research team, or by data contributors themselves using the GGIR package processing script provided by the core research team. Reprocessed individual study datasets were then pooled into a single dataset. A detailed description of the reprocessing method and decisions for specific accelerometer devices is available in an online supplementary file.

For sleep, GGIR produced the following metrics; total sleep time (minutes), sleep efficiency (percentage of time asleep within the sleep period), sleep regularity (regularity of sleep and awake patterns on a day-to-day basis calculated as the sleep regularity index), and timing (time of sleep and awakening within the 24-hour time cycle). For physical activity, we used volume and intensity metrics proposed by Rowlands et al [35]. These measures are 1) average acceleration (measure of physical activity volume) and 2) intensity gradient (measure of intensity distribution where the natural logs of time and intensity are used to produce a straight-line graph with the slope of the graph being the intensity gradient). We used this approach to model the full continuum of physical activity intensity for our main analyses [36] and avoid collapsing physical activity data into population-specific intensity categories which occurs when intensity cut-points are applied to accelerometer data [37].

For the purposes of translating physical activity data for interpretation and public health recommendations and for time-use based analyses we then determined; 1) the number of minutes spent in acceleration intensity bands in 50mg increments [35], 2) the number of minutes spent in light and moderate to vigorous physical activity using age appropriate cut-points applied to the time in intensity bands data, and 3) the minimum acceleration measured for a range of different durations, this is the MX metric where X refers to the duration. For example, $M60$ refers to the minimum acceleration for the most active 60 minutes of the day [36]. We also categorise all awake time into bout lengths for sedentary, light, and moderate to vigorous physical activity bouts using the categories defined by Gába et al [38]. After extracting sleep and physical activity, the remaining time within the 24-hour time cycle was classified as sedentary behaviour. Non-wear time and abnormally high acceleration values were detected using the data processing default setting [39]. These data were classified as missing and imputed using the accelerometer data from the same time interval on the remaining days of the week.

Covariates and moderators

The following individual participant data were obtained when available; age, sex, height, weight, waist circumference, maturational status, socioeconomic status (SES), ethnicity, screen time, sleep medications, and medically diagnosed conditions known to affect sleep. The country, city, time zone, daylight length, and season in which data were collected were also obtained.

Statistical analysis

To reduce the risk of bias in secondary data analysis of IPD, we used a blind analysis approach [40]. We developed and tested models on a 10% holdout subsample of the dataset. When the analysis code was ready, we conducted the final analysis [41].

We checked all data for implausibly low or high measurements which were excluded if they fell outside of ±4SD of the mean, as these likely indicate measurement device failures. We assumed missing covariate and moderator data to be missing at random or completely at random and imputed these missing values using multiple imputation by chained equations with the *mice* package in R [42] to provide 50 imputed datasets. Specifically, we used multilevel predictive mean matching imputation clustered by participants. We used Rubin's rule to pool the results from analyses conducted using each imputed dataset into one single set of results. For all analyses, we examined 95% confidence intervals and used $P < 0.05$ to assess for statistical significance. We used R (version 4.3.2) for all data cleaning and analysis. We report mean and 95% confidence intervals. For categorical variables, percentages are reported. We examined model diagnostics from all models (see supplementary materials).

To examine the relationship between sleep and physical activity (Research Questions 1-2) we fit mixed-effects models with random intercepts for study ID to account for the nesting of participants in studies [43]. We predict each sleep indicator by each physical activity indicator controlling for sex, body mass index (BMI) and region. Quadratic terms were introduced in each model to allow for curvilinear relationships to be modelled. We interacted each predictor with age to reveal how the relationship between physical activity and sleep changes over the lifespan.

Protocol deviations

This paper addresses Hypotheses 1 and 2 from our registered protocol. We will address remaining hypotheses in future research. To streamline our analysis and reduce redundancy, we made some changes to our protocol while working with the holdout data.

Using Study ID as fixed effect. We initially planned to use Study ID as a fixed effect, given that our study selection was not random. But, as we aimed for our findings to generalise to other studies and populations, we instead opted to include random intercepts for Study ID which can better model generalisable patterns. This approach allows us to account for variability between studies without fully partialing out study effects, thereby better modelling common phenomena. While this change lessened the strength of model intercepts, it did not significantly alter model slopes. The results using Study ID as a fixed effect (i.e., those that align with the protocol) can be found in the supplementary materials, and are qualitatively similar to the main results.

Sleep indicator variable. Our research examines various metrics of sleep: onset, duration, regularity, and efficiency. We initially planned to merge these variables into a single indicator to simplify analysis. However, no validated indicator existed which could be estimated from our measured variables. We decided against creating one, as we could not validate it with our data. Consequently, our models illustrate the relationship between each aspect of sleep and physical activity, instead of describing a straightforward link between overall sleep quality and physical activity.

Log transformation. We aimed to improve model diagnostics using log transformations of skewed variables. Pilot analysis using holdout data revealed that while diagnostics improved on the log scale, model predictions followed extreme exponential patterns near the edge of the data range when back-transformed. Accordingly, we do not use log transformations in the main paper, however these models can be found in supplementary materials.

Patient and public involvement

Patients and members of the public were not involved in the design, analysis or interpretation of this IPD study using secondary data. Results of this study have broad public health interest, particularly policymakers. Findings from this study will be disseminated through institutional websites, press and media platform releases, and tailored messages to public health organisations and governing bodies.

Results

The aggregated data-set describes 220,612 observations of daily physical activity and sleep from 28,823 unique participants. Table 1 shows demographic information for all participants. Observations were not uniform across the days of the week $\chi^2_{(6)} = 80.75$, p = $< .001$. There were fewer observations on Monday (z = -7.80) and Sunday (z = -3.01); and more on Wednesday (z = 2.98). A table of study characteristics can be found in supplementary materials.

The effects of physical activity volume on sleep

We estimated the effects of physical activity on sleep (RQ1) using mixed-effects models. The effect of physical activity volume on sleep by age are presented in Table 2 and Figure 1. Higher physical activity volume was associated with longer sleep duration, higher sleep efficiency, earlier sleep onset, and more regular sleep. While we observed statistically significant curvilinear relationships, these did not appear to meaningfully change the effect. The relationship between physical activity volume and sleep duration, efficiency, and onset was consistent across the age groups, with some very weak evidence that the relationship with sleep regularity grew stronger with age.

The effects of physical activity intensity on sleep

We estimated how physical activity intensity affects sleep across different age groups and present the results controlling for sex, SES, and BMI, in Table 2 and Figure 2. We observed weak associations between physical activity intensity and sleep duration, sleep efficiency, and sleep onset, with a stronger relationship observed with sleep regularity. As with physical activity volume, we observed statistically significant curvilinear relationships, but these were only meaningful for the impact of physical activity intensity on sleep regularity, in which very low levels of intensity were strongly associated with poor sleep regularity and the benefits of increasing intensity diminished past one standard deviation above the mean. While age statistically significantly modified the relationship between physical activity intensity and sleep duration, efficiency, and regularity, the effect was negligible for duration and efficiency. For regularity, the association weakened with age.

The effects of sleep duration on physical activity

We estimated the effect of sleep duration on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 3. As age increases, both physical activity volume and intensity decrease. We found no evidence for an association between average sleep duration and physical activity volume.

The effects of sleep efficiency on physical activity

We estimated the effect of sleep efficiency on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 4. There was a negligible relationship between sleep efficiency and physical activity volume and a weak linear association between sleep efficiency and more intense physical activity.

The effects of sleep onset on physical activity

We estimated the effect of sleep onset on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 5. There were weak curvilinear relationships where average sleep onset was linked to the highest levels of physical activity volume, but not intensity.

The effects of sleep regularity on physical activity

We estimated the effect of sleep regularity on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 6. There was a positive linear relationship between sleep regularity and both physical activity volume and intensity. These relationships were slightly attenuated with age, such that for older adults the effect was negligible.

Discussion

In this study, we collated and harmonised device-measured physical activity and sleep data from 20 studies around the world, covering participants across the lifespan. We found that both the amount and the intensity of physical activity was associated with longer sleep duration, higher sleep efficiency, earlier sleep onset, and more regular sleep. These relationships were the strongest for sleep regularity, although the the relationship between physical activity intensity weakened with age. Many of these relationships were bi-directional.

Sleep was more regular among people who engaged in more intense physical activity. Further, we found no evidence that sleep duration or efficiency were associated with physical activity volume. Those who slept more efficiently engaged in more intense physical activity the following day. Those who fell asleep earlier and at a more regular time also engaged in more intense physical activity, and engaged in more total physical activity. These findings were largely robust to the decisions made during analysis (see supplementary materials for all analyses).

The lack of association between physical activity and sleep duration or efficiency is particularly notable, as a relationship between these variables has been observed in meta-analyses [8]. For example, [9] found small ($0.22 < d < 0.30$) but significant relationships between physical activity and sleep duration and efficiency in adults using experimental studies for both acute and regular physical activity. This discrepancy could be due to the design of the studies, as the majority of studies which have found a relationship were experimental. That is, physical activity was manipulated and it may be that the routine nature of individuals' physical activity makes associations more difficult to detect. Similarly, most of the studies which examined acute changes to physical activity were conducted in laboratory settings. These studies, while useful for understanding the potential mechanisms, are not ecologically valid and may not reflect the relationship between physical activity and sleep in the real world. Laboratory studies also tend to use more sensitive measures of sleep and physical activity, which may generate less statistical noise and make relationships easier to detect. It could also be that duration of sleep is set more by parameters external to the individual, such as work schedules or parenting duties, than by individual health behaviours.

We found the most consistent evidence for relationships between sleep onset and regularity with physical activity volume and intensity. This relationship appeared to be bi-directional, with sleep onset and regularity influencing physical activity the following day, and physical activity influencing sleep onset and regularity that night. This is consistent with the idea that sleep and physical activity are mutually reinforcing behaviours [8], or that they are habitual and linked to factors such as the value placed on health. Sleep regularity is a stronger predictor of mortality than sleep duration [44], and so the finding that physical activity is associated with sleep regularity may be particularly important.

Interestingly, the effects of physical activity on sleep regularity were stronger in older adults compared to younger individuals. This age-dependent association suggests that promoting physical activity in older adults could be a particularly effective strategy for improving sleep health, which may contribute to better overall health outcomes [2,3]. Alternatively, it could mean that older adults have fewer of the external constraints (e.g., work schedules, parenting duties) that younger adults have, and so are more able to benefit from the effects of physical activity on sleep. Furthermore, the age-related weakening of the positive effects of physical activity volume beyond a certain threshold implies that moderate rather than high levels of physical activity may be sufficient for promoting sleep in older populations. For younger individuals, physical activity was associated with earlier sleep onset, whereas for older participants, physical activity was linked to later sleep onset. This suggests a nuanced relationship, where age-specific factors (such as changes in circadian rhythm, or age-varying commitments such as work or school) might mediate how physical activity impacts sleep timing. Future public health recommendations could consider tailoring activity guidelines to different age groups to optimise sleep benefits.

These findings have other important implications for public health recommendations. Our results support encouraging physical activity for maintaining regular sleep patterns, especially in older adults. However, the data suggest that focusing solely on increasing physical activity volume or intensity may not necessarily improve other aspects of sleep quality, such as duration or efficiency. Public health messages should, therefore, emphasise the role of physical activity in promoting stable sleep schedules rather than framing physical activity as a universal solution for improving sleep. We note, however, that we did not examine the effects of physical activity on sleep in individuals with sleep disorders or other health conditions. Physical activity interventions may still be a useful treatment for these conditions.

Our study has several strengths, including the use of device-based measures, which reduce the bias associated with self-reported sleep and activity data, and the harmonisation of individual data from multiple studies, conducted in geographically diverse regions and among multiple age groups, allowing for robust and generalisable findings. However, the study is not without limitations. Despite our efforts to harmonise data, variability in accelerometer wear locations and slight differences in data collection protocols across studies may have introduced measurement inconsistencies. Additionally, although we controlled for several potential confounders, there may still be unmeasured factors affecting the observed associations. Finally, while our study captured data from across the lifespan, the limited representation of certain age groups, especially adolescents and young adults, restricts the generalisability of our conclusions to these populations. The limited research on those aged 18-35 is particularly concerning given the high prevalence of poor sleep in this age group [7].

Conclusion

Our findings contribute to the growing body of literature that highlights the complex relationship between physical activity and sleep. While physical activity appears to benefit sleep regularity, its effects on other sleep metrics are less clear, particularly when considering age differences. Future research should aim to explore mechanisms underlying these associations and identify specific intervention strategies that can effectively enhance both sleep and physical activity behaviours for diverse populations.

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Table 1

Participant characteristics

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Numeric variables					
N	9,758	3,062	473	4,675	6,563
Age	9.49 (1.51)	13.71 (2.39)	23.90 (6.13)	56.72 (7.55)	72.21 (4.73)
BMI	17.95 (3.30)	20.39 (4.00)	24.71 (5.88)	26.87 (4.95)	27.08 (4.42)
Valid weartime hours	22.18 (2.35)	22.18 (2.67)	22.15 (2.63)	22.75 (2.23)	23.82 (0.75)
PA intensity	-2.12 (0.19)	-2.20 (0.19)	-2.37 (0.19)	-2.49 (0.20)	-2.67 (0.22)
PA volume	43.68 (25.61)	42.87 (16.63)	40.64 (11.15)	39.56 (13.10)	31.27 (10.38)
Sleep duration	446.73 (92.90)	403.02 (96.97)	365.62 (79.72)	400.68 (63.47)	398.74 (71.46)
Sleep efficiency	0.79 (0.10)	0.83 (0.10)	0.87 (0.07)	0.88 (0.06)	0.86 (0.06)
Sleep onset	20.96 (1.17)	22.30 (1.96)	24.00 (1.47)	23.65 (1.35)	24.41 (1.30)
Sleep regularity	54.54 (12.67)	54.12 (13.05)	54.75 (12.83)	59.41 (11.52)	54.73 (12.55)
Accelerometer Wear Location					
Wrist	2,479 (25.40%)	1,928 (62.97%)	473 (100.00%)	4,675 (100.00%)	6,563 (100.00%)
Hip	7,279 (74.60%)	1,134 (37.03%)	-	-	-
Ethnicity					
Non-white	1,830 (18.75%)	469 (15.32%)	34 (7.19%)	87 (1.86%)	263 (4.01%)
Unclear	4,016 (41.16%)	1,620 (52.91%)	235 (49.68%)	1,317 (28.17%)	2,475 (37.71%)
White	3,912 (40.09%)	973 (31.78%)	204 (43.13%)	3,271 (69.97%)	3,825 (58.28%)
Region					
Europe	3,203 (32.82%)	1,189 (38.83%)	234 (49.47%)	4,305 (92.09%)	6,560 (99.95%)
Oceania	1,761 (18.05%)	552 (18.03%)	123 (26.00%)	370 (7.91%)	3 (0.05%)
South america	2,074 (21.25%)	984 (32.14%)	116 (24.52%)	-	-

Table 1 continued

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Africa	876 (8.98%)	167 (5.45%)	-	-	-
Asia	610 (6.25%)	90 (2.94%)	-	-	-
North america	1,234 (12.65%)	80 (2.61%)	-	-	-
Season					
Autumn	3,183 (32.62%)	663 (21.65%)	54 (11.42%)	1,149 (24.58%)	1,404 (21.39%)
Spring	2,263 (23.19%)	1,599 (52.22%)	264 (55.81%)	1,156 (24.73%)	2,128 (32.42%)
Summer	1,223 (12.53%)	314 (10.25%)	86 (18.18%)	1,175 (25.13%)	1,358 (20.69%)
Winter	3,089 (31.66%)	486 (15.87%)	69 (14.59%)	1,195 (25.56%)	1,673 (25.49%)
Sex					
Female	5,118 (52.45%)	1,593 (52.02%)	306 (64.69%)	2,619 (56.02%)	2,725 (41.52%)
Male	4,640 (47.55%)	1,469 (47.98%)	167 (35.31%)	2,056 (43.98%)	3,838 (58.48%)
Sleep Conditions Reported					
Yes	49 (0.50%)	1 (0.03%)	-	261 (5.58%)	682 (10.39%)
Socioeconomic Status					
Low	3,607 (36.96%)	819 (26.75%)	65 (13.74%)	1,162 (24.86%)	2,490 (37.94%)
Medium	2,670 (27.36%)	1,105 (36.09%)	265 (56.03%)	2,221 (47.51%)	2,569 (39.14%)
High	3,481 (35.67%)	1,138 (37.17%)	143 (30.23%)	1,292 (27.64%)	1,504 (22.92%)

Note. 88 participants had missing age data. For categorical variables the value is the count, and percentage. For numeric variables the value is the Mean and SD. N = 28,823

Table 2

Physical activity predicting sleep controlling for SES, sex, and BMI.

	Physical activity volume (z)				Physical activity intensity (z)			
Term	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
Sleep duration (z)								
(Intercept)	0.24 [0.06, 0.41]	0.09	2.69	.007	0.24 [0.07, 0.40]	0.08	2.85	.004
Physical activity	0.10 [0.08, 0.12]	0.01	10.88	$< .001$	0.08 [0.07, 0.10]	0.01	10.50	$< .001$
Age	0.00 [0.00, 0.00]	0.00	-3.00	.003	0.00 [0.00, 0.00]	0.00	-2.92	.004
Physical activity^2	-0.01 [-0.01, 0.00]	0.00	-4.94	$< .001$	0.01 [0.00, 0.02]	0.00	2.76	.006
Physical activity × age	0.00 [0.00, 0.00]	0.00	-3.57	$< .001$	0.00 [0.00, 0.00]	0.00	-5.73	$< .001$
Age × Physical activity^2	0.00 [0.00, 0.00]	0.00	-0.05	.959	0.00 [0.00, 0.00]	0.00	-4.89	$< .001$
Sleep efficiency (z)								
(Intercept)	0.26 [0.09, 0.44]	0.09	2.92	.004	0.29 [0.10, 0.47]	0.10	2.99	.003
Physical activity	0.10 [0.09, 0.12]	0.01	11.30	$< .001$	0.06 [0.05, 0.08]	0.01	7.97	$< .001$
Age	0.00 [0.00, 0.00]	0.00	-0.18	.855	0.00 [0.00, 0.00]	0.00	-0.86	.391
Physical activity^2	-0.02 [-0.02, -0.02]	0.00	-11.97	$< .001$	-0.03 [-0.04, -0.02]	0.00	-6.69	$< .001$
Physical activity × age	0.00 [0.00, 0.00]	0.00	-10.55	$< .001$	0.00 [0.00, 0.00]	0.00	-7.17	$< .001$
Age × Physical activity^2	0.00 [0.00, 0.00]	0.00	4.79	$< .001$	0.00 [0.00, 0.00]	0.00	5.40	$< .001$
Sleep onset (z)								
(Intercept)	0.00 [-0.27, 0.27]	0.14	0.01	.995	0.00 [-0.26, 0.26]	0.13	0.01	.988
Physical activity	-0.03 [-0.04, -0.01]	0.01	-3.61	$< .001$	-0.06 [-0.07, -0.05]	0.01	-10.30	$< .001$
Age	0.00 [0.00, 0.00]	0.00	-1.73	.083	0.00 [0.00, 0.00]	0.00	-1.22	.222
Physical activity^2	-0.01 [-0.01, 0.00]	0.00	-4.30	$< .001$	-0.03 [-0.04, -0.02]	0.00	-9.25	$< .001$
Physical activity × age	0.00 [0.00, 0.00]	0.00	-11.48	$< .001$	0.00 [0.00, 0.00]	0.00	1.35	.178
Age × Physical activity^2	0.00 [0.00, 0.00]	0.00	8.83	$< .001$	0.00 [0.00, 0.00]	0.00	11.29	$< .001$
Sleep regularity (z)								

Table 2 continued

	Physical activity volume (z)				Physical activity intensity (z)			
Term	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
(Intercept)	0.53 [0.38, 0.67]	0.07	7.18	$<$ .001	0.71 [0.53, 0.89]	0.09	7.60	$<$ .001
Physical activity	0.24 [0.22, 0.25]	0.01	25.42	$<$ .001	0.27 [0.25, 0.29]	0.01	29.15	$<$ .001
Age	0.00 [0.00, 0.00]	0.00	1.06	.293	0.00 [-0.01, 0.00]	0.00	-3.08	.003
Physical activity^2	-0.03 [-0.03, -0.03]	0.00	-15.45	$<$ .001	-0.10 [-0.11, -0.09]	0.01	-18.83	$<$ .001
Physical activity $\times$ age	0.00 [0.00, 0.00]	0.00	-1.48	.141	0.00 [0.00, 0.00]	0.00	-21.58	$<$ .001
Age $\times$ Physical activity^2	0.00 [0.00, 0.00]	0.00	-5.89	$<$ .001	0.00 [0.00, 0.00]	0.00	9.47	$<$ .001

Note. Adjusted for SES, sex, and BMI. Outcomes variables are listed in the column headers.

Table 3

Sleep predicting physical activity controlling for SES, sex, and BMI

	Physical activity volume (z)				Physical activity intensity (z)			
Term	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
Sleep duration (z)								
(Intercept)	1.64 [1.29, 1.99]	0.18	9.17	< .001	1.46 [1.32, 1.61]	0.07	20.34	< .001
Sleep duration	0.00 [-0.02, 0.03]	0.01	0.24	.813	0.00 [-0.01, 0.02]	0.01	0.43	.667
Age	-0.03 [-0.03, -0.02]	0.00	-38.29	< .001	-0.03 [-0.03, -0.03]	0.00	-48.38	< .001
Sleep duration^2	0.00 [-0.01, 0.00]	0.00	-1.41	.161	0.00 [-0.01, 0.00]	0.00	-0.70	.486
Sleep duration × age	0.00 [0.00, 0.00]	0.00	-1.34	.184	0.00 [0.00, 0.00]	0.00	1.96	.052
Age × Sleep duration^2	0.00 [0.00, 0.00]	0.00	-3.23	.002	0.00 [0.00, 0.00]	0.00	-2.15	.032
Sleep efficiency (z)								
(Intercept)	1.63 [1.28, 1.98]	0.18	9.13	< .001	1.46 [1.32, 1.61]	0.07	20.40	< .001
Sleep efficiency	0.02 [0.00, 0.03]	0.01	1.98	.051	0.03 [0.02, 0.04]	0.01	4.45	< .001
Age	-0.03 [-0.03, -0.02]	0.00	-38.52	< .001	-0.03 [-0.03, -0.03]	0.00	-48.47	< .001
Sleep efficiency^2	0.00 [0.00, 0.01]	0.00	1.86	.065	0.00 [0.00, 0.01]	0.00	4.28	< .001
Sleep efficiency × age	0.00 [0.00, 0.00]	0.00	-0.89	.373	0.00 [0.00, 0.00]	0.00	-4.13	< .001
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	0.00	-1.32	.188	0.00 [0.00, 0.00]	0.00	-2.79	.005
Sleep onset (z)								
(Intercept)	1.65 [1.29, 2.00]	0.18	9.12	< .001	1.47 [1.33, 1.61]	0.07	20.28	< .001
Sleep onset	-0.03 [-0.08, 0.01]	0.02	-1.41	.165	-0.01 [-0.04, 0.02]	0.02	-0.48	.630
Age	-0.03 [-0.03, -0.02]	0.00	-38.31	< .001	-0.03 [-0.03, -0.03]	0.00	-48.15	< .001
Sleep onset^2	-0.02 [-0.03, 0.00]	0.01	-2.61	.011	0.00 [-0.01, 0.01]	0.00	0.01	.992
Sleep onset × age	0.00 [0.00, 0.00]	0.00	1.16	.249	0.00 [0.00, 0.00]	0.00	0.82	.414
Age × Sleep onset^2	0.00 [0.00, 0.00]	0.00	1.34	.182	0.00 [0.00, 0.00]	0.00	-0.78	.435
Sleep regularity (z)								

Table 3 continued

	Physical activity volume (z)				Physical activity intensity (z)			
Term	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
(Intercept)	1.59 [1.25, 1.94]	0.18	9.08	< .001	1.45 [1.31, 1.58]	0.07	20.69	< .001
Sleep regularity	0.11 [0.09, 0.12]	0.01	12.20	< .001	0.08 [0.07, 0.09]	0.01	13.75	< .001
Age	-0.02 [-0.03, -0.02]	0.00	-38.26	< .001	-0.03 [-0.03, -0.03]	0.00	-48.34	< .001
Sleep regularity^2	0.00 [-0.01, 0.00]	0.00	-0.43	.668	0.00 [-0.01, 0.00]	0.00	-2.11	.035
Sleep regularity $\times$ age	0.00 [0.00, 0.00]	0.00	-10.39	< .001	0.00 [0.00, 0.00]	0.00	-12.13	< .001
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	0.00	-0.05	.959	0.00 [0.00, 0.00]	0.00	0.53	.598

Note. Adjusted for SES, sex, and BMI. Outcomes variables are listed in the row headers.

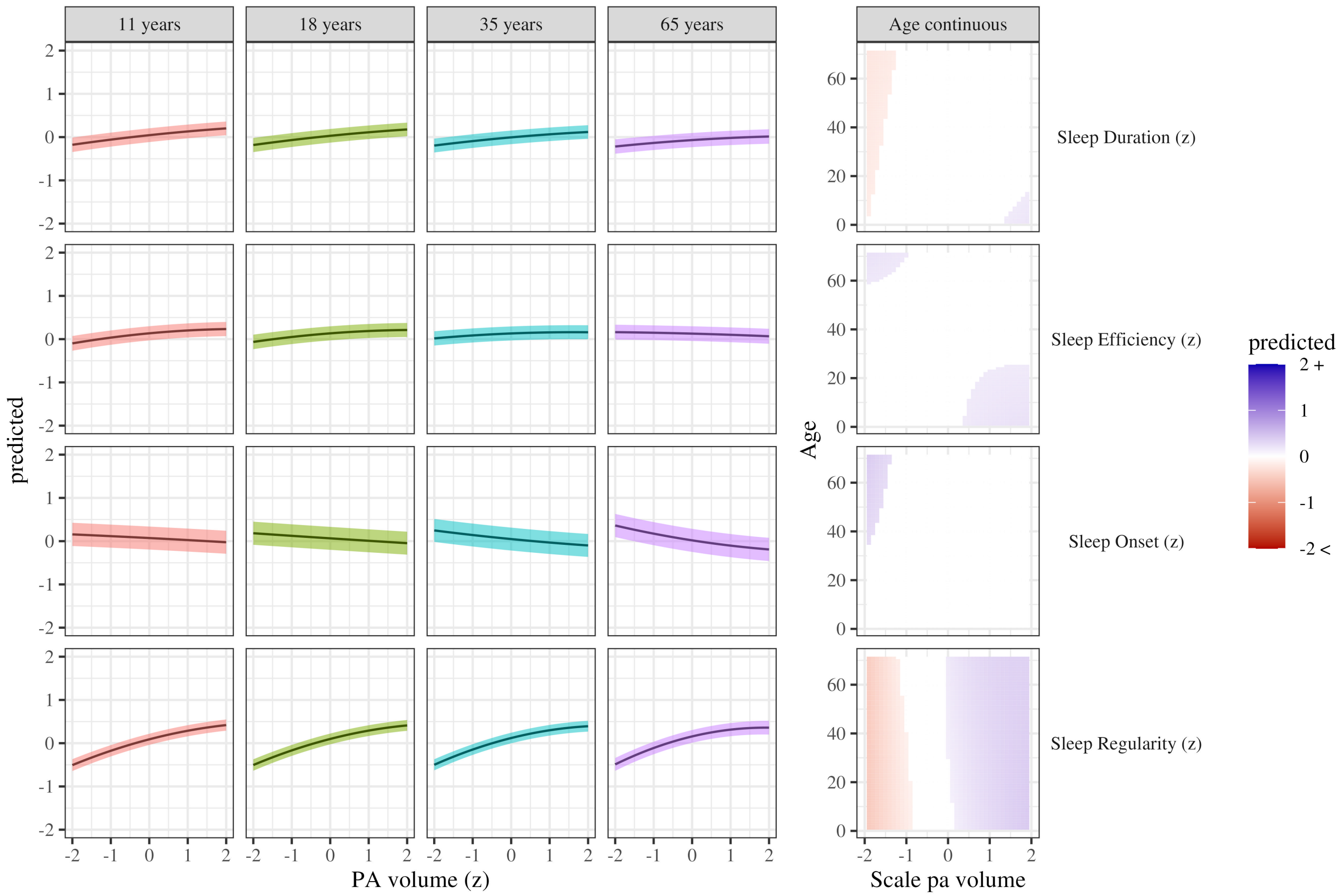

Figure 1. Sleep metrics on physical activity volume. The panels on the left show the curvilinear relationship between PA volume and each of the sleep outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero. These plots demonstrate exact turning points where the effects of PA volume change by age.

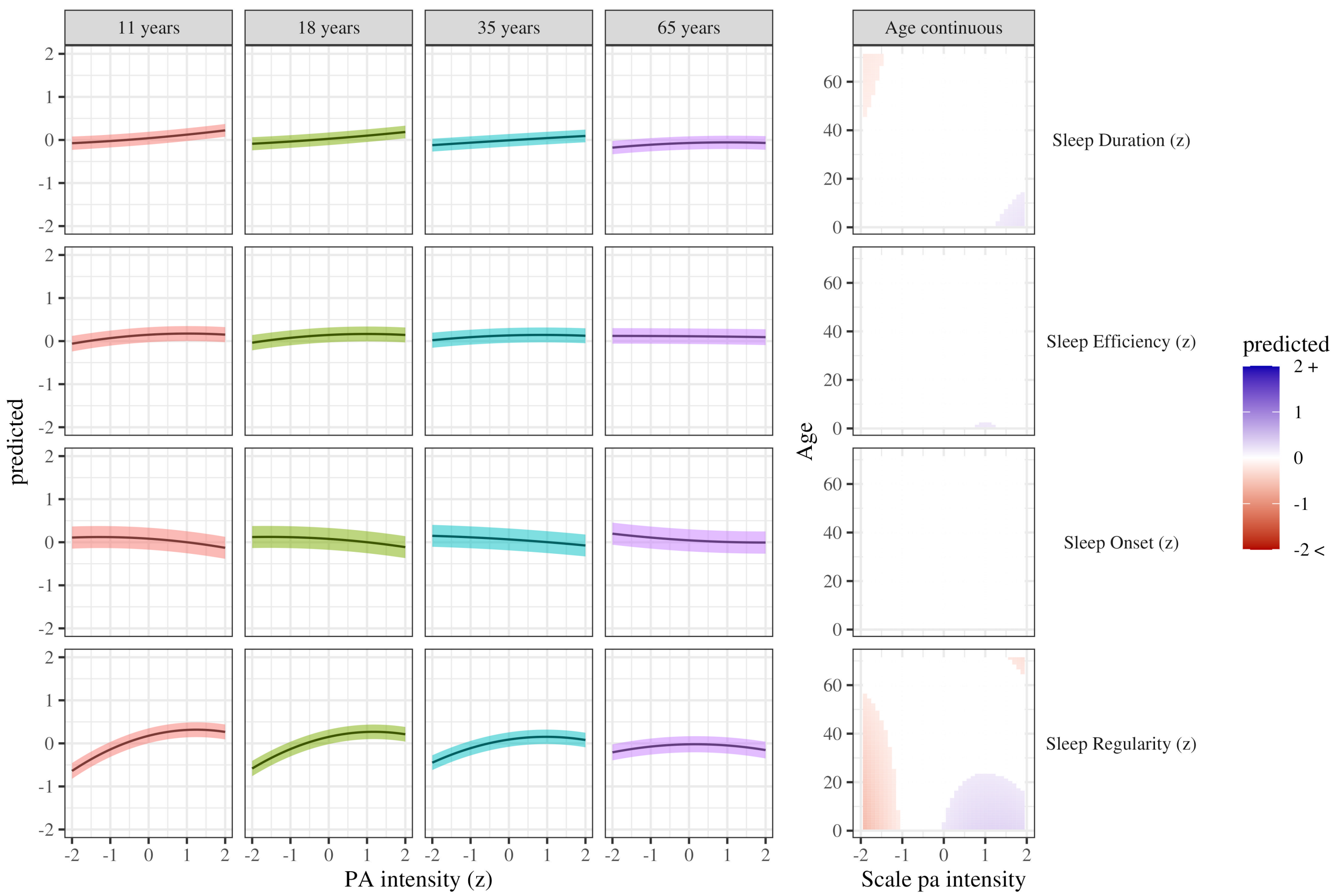

Figure 2. Sleep metrics on physical activity intensity. The panels on the left show the curvilinear relationship between PA intensity and each of the sleep outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

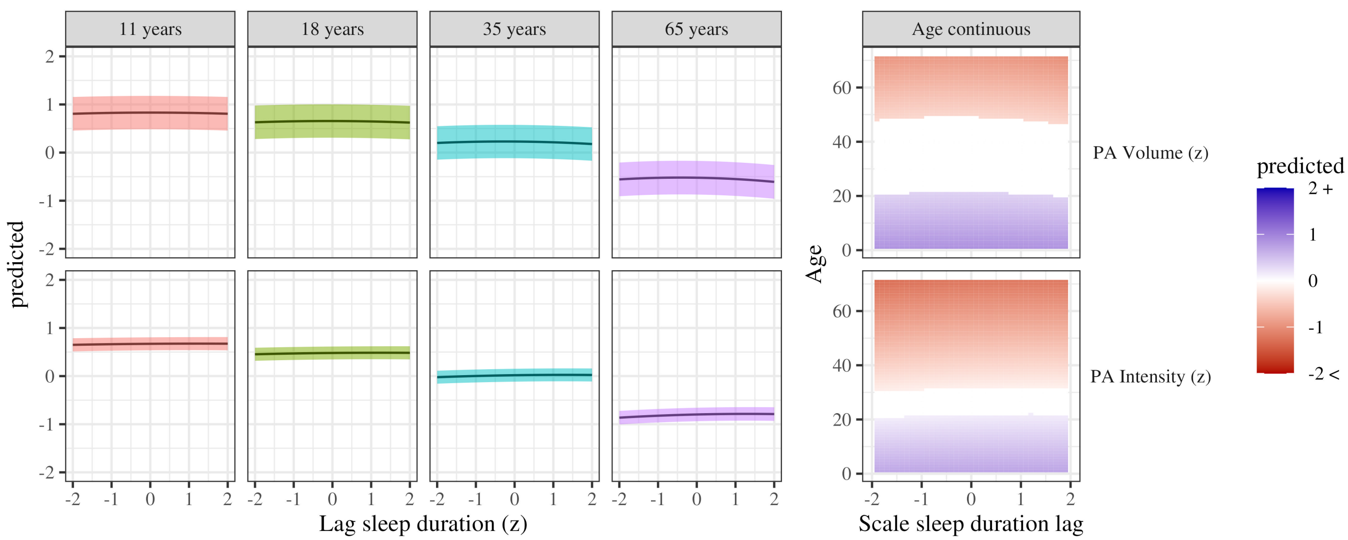

Figure 3. Physical activity by sleep duration. The panels on the left show the curvilinear relationship between sleep duration and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

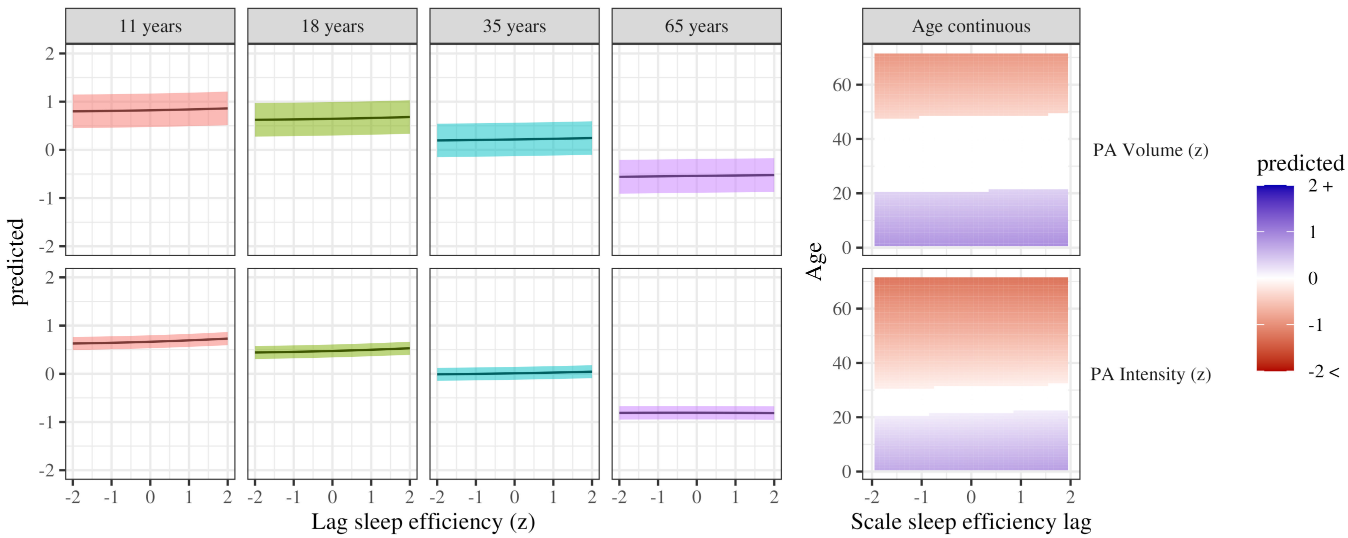

Figure 4. Physical activity by sleep efficiency. The panels on the left show the curvilinear relationship between sleep efficiency and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

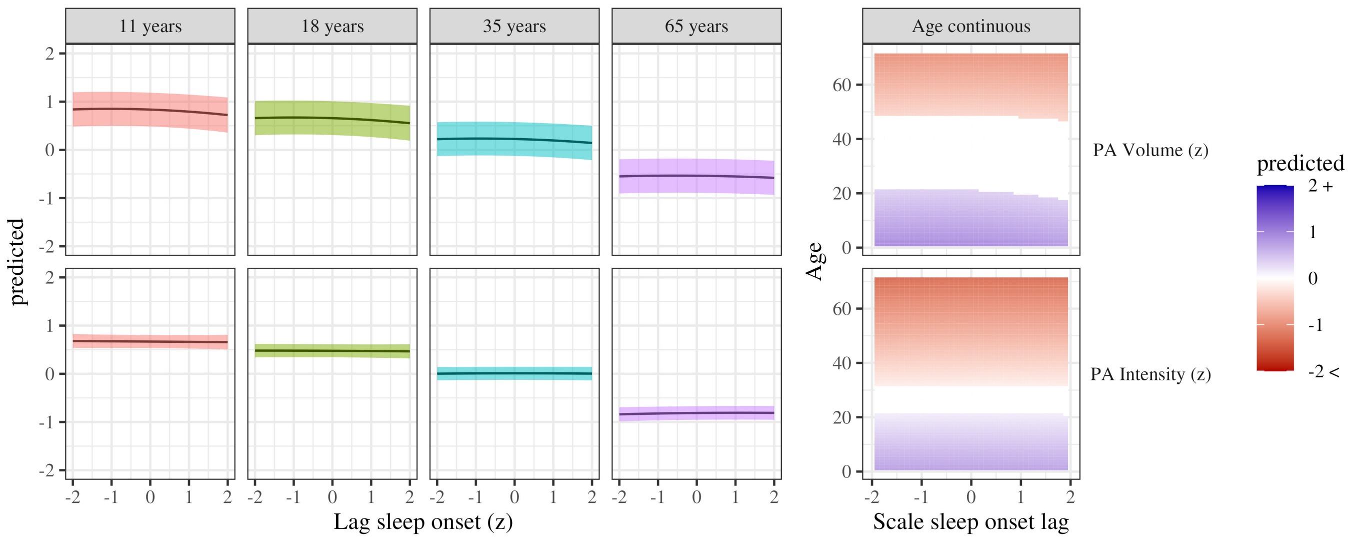

Figure 5. Physical activity by sleep onset. The panels on the left show the curvilinear relationship between sleep onset and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

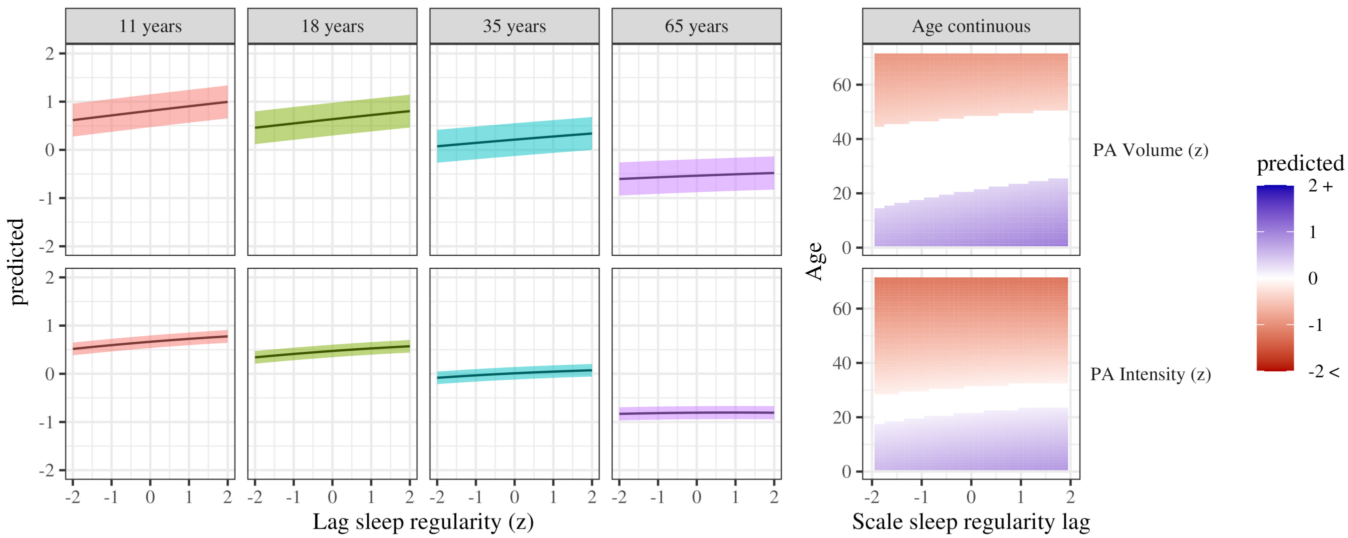

Figure 6. Physical activity by sleep regularity. The panels on the left show the curvilinear relationship between sleep regularity and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.